CM0832 Elements of Set Theory
8. Ordinals and Order Types

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Preliminaries

Textbook

Herbert B. Enderton (1977). Elements of Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Ordinal Addition

Definition

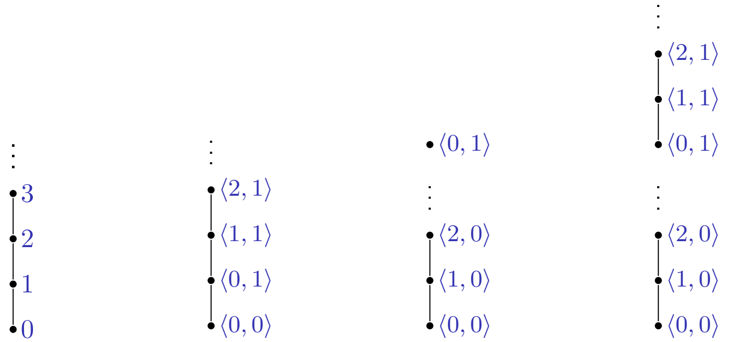
Let α and β be ordinals. We define their **addition** by transfinite recursion on β:

$$\begin{aligned}
\alpha + 0 &= \alpha, \\
\alpha + \beta^+ &= (\alpha + \beta)^+, \\
\alpha + \beta &= \sup \{ \alpha + \lambda \mid \lambda < \beta \} \\
&= \bigcup_{\lambda < \beta} (\alpha + \lambda), \text{ for all limit ordinal } \beta.
\end{aligned}$$

Ordinal Addition

Example

$\omega = 1 + \omega$, $1 + \omega \neq \omega + 1$ and $\omega + 1 \neq \omega + \omega$.



(a) ω (b) $1 + \omega$ (c) $\omega + 1$ (d) $\omega + \omega$

References

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 2).